

Collapse as the Present: George Ellis’s Evolving Block Universe and Event Density’s Committing Substrate

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Preamble — what this article does and does not claim

1. **This is a comparison and position paper, not a new derivation.** It maps an independent, established research program — George Ellis’s Evolving Block Universe (EBU) and its quantum refinement, the Crystallizing Block Universe (CBU) — onto Event Density, and identifies where each supplies what the other lacks. Ellis’s results are *cited, not re-derived*; ED’s are its standing conditional derivations, cited inline.
 2. **Convergence, not endorsement.** Ellis has, to the author’s knowledge, no acquaintance with Event Density. Every point of agreement below is **independent arrival at the same conclusion from a different starting point** (Ellis from general relativity and the measurement problem; ED from a discrete substrate), never Ellis’s sanction of ED. The phrase used throughout is *convergence*.
 3. **ED reproduces the physics both frameworks share.** Nothing here revises quantum mechanics or general relativity. The ED-specific content invoked (the constructed substrate, the collapse scale M_{crit} , the dissolution of the Past Hypothesis) carries its own standing tier, flagged per claim.
 4. **The two things ED offers Ellis’s program are tiered, not asserted as fact.** (a) A *constructed* substrate whose commitment drives the block’s growth is a conditional structural derivation from the thirteen ED primitives (Paper_087), not a completed quantum-gravity theory. (b) A collapse mass-scale M_{crit} is an *unconfirmed prediction* (Paper_060), not a result. Neither closes Ellis’s open problems; each is a candidate answer to one he names.
 5. **Tiers marked per claim.** The core convergence (collapse-as-present $\approx$ commitment) is *interpretive and structural*; the substrate engine is a *conditional derivation*; M_{crit} is a *prediction*; the Past-Hypothesis dissolution is a *reframe* (Paper_HowTheoryCoarseGrainReality §13); the preferred-frame comparison is a *limited structural convergence* — both reject Minkowski realism and carry preferred worldlines, but only ED is in the khronometric modified-gravity class, and the two disagree on the source of the frame (§6).
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1. What Event Density is, and the class this runs in

Event Density is a discrete, relational substrate: a participation graph whose channels carry a bandwidth $b \geq 0$ and a phase, with one process primitive — **commitment** (P11), the irreversible act by which an uncommitted participation becomes a definite, recorded fact, writing the arrow of time into the law rather than onto it (Paper_087). Space and geometry are emergent: the metric is the reading of the bandwidth field, $g \sim 1/b$ (Paper_MetricFromTheGraph, Paper_GR-I). Measurement is not a separate postulate — it *is* commitment: the basis-selecting, irreversible actualization of one outcome from a superposed participation (Report §4). ED’s gravitational class is **khronometric**: two tensor modes plus one propagating scalar, the *khronon* $u_\mu = \partial_\mu T/|\partial T|$, the arrow made geometric (Paper_GR-II), with luminal cones collapsing the coupling space to one dimension λ_{local} , on which $\alpha_2 = 0$ exactly and $\alpha_1 = -4\lambda_{local}$, suppressed ~ 70 orders below current bounds (Paper_GR-IV).

Held in that frame, ED makes four claims that will matter here: (i) the present is the live edge where the uncommitted becomes committed; (ii) the future is genuinely open (uncommitted) and the past fixed (recorded); (iii) the arrow is primitive, and the smooth time-reversible laws are its coarse-grained shadow (Paper_HowTheoryCoarseGrainReality); (iv) collapse is real, basis-selecting, and irreducible to unitary evolution.

Every one of those four is also a load-bearing claim of Ellis’s EBU/CBU — reached from the opposite direction.

2. Ellis’s Evolving and Crystallizing Block Universe

Ellis (2006–2014; *On the Flow of Time*; *Time and Spacetime: The Crystallizing Block Universe*, with Rothman; *The Evolving Block Universe and the Meshing Together of Times*) rejects both the static block (eternalism, “time is an illusion”) and presentism. His ontology is a **growing** block: the past and present exist and are fixed; the **future does not exist** — it is a set of possibilities, not a region. Spacetime is the standard four-dimensional block *minus the future*, with a moving growth-front. The present is the growing edge, and it is ontologically special: it is where the indefinite future becomes the definite past.

The mechanism is the paper’s title-phrase: **collapse is the present**. In Ellis’s own italic hypothesis, the uncertainty of the future changes to the certainty of the past when wave-function collapse occurs — “all the time, everywhere.” Collapse is not a measurement *problem*; it is time happening. This is an **objective-collapse** position: Ellis treats reduction as a real, time-irreversible physical process, extra to unitary quantum theory, and explicitly rejects the three standard evasions — the ensemble reading (no ensemble without individual events), decoherence-as-sufficient (it diagonalizes but does not select one outcome), and many-worlds (no block, plus the measure and preferred-basis problems). The future is open **ontologically**, from real quantum indeterminism amplified to macro and cosmic scale (inflation, chaos), not merely epistemically.

The CBU is the quantum refinement: because of delayed-choice and quantum-eraser evidence, the transition from indefinite to definite does not occur on a clean simultaneity surface. The crystallization surface is *crinkly and lagging* — some micro “past” events fix only later — so “the past is fixed” is strictly the macroscopic, coarse-grained statement.

Two further commitments are essential for the comparison, and both are exactly where ED has

something to add:

- **The deep substrate is posited, not built.** Ellis holds that classical spacetime and its growth must emerge from an underlying quantum-gravity layer, that the growth is “*presumably*” collapse of a quantum-gravity wave function, and that any correct quantum-gravity theory *must* yield an EBU at the macro level — a selection rule. He notes discrete programs (causal sets, spin foams) as compatible. He does not construct the layer.
- **A theory of collapse is named as the missing piece.** Ellis states that the outstanding major issue for the program is a viable theory of *when and where* collapse occurs; he does **not** adopt a Penrose–Diosi mass threshold, and treats the collapse scale as set contextually by the local environment, with no universal number.

3. The convergence: collapse-as-present $\approx$ commitment

The central result of this comparison is a near-identity of the two frameworks’ account of *now*, reached independently.

	Ellis (EBU/CBU)	Event Density
The present	the growing edge; indefinite → definite	the live edge; uncommitted → committed (P11)
“Now” mechanism	objective wave-function collapse <i>is</i> the flow of time	commitment <i>is</i> the arrow (P11)
Future / past	future does not exist (open); past fixed/recorded	future uncommitted (open); past committed/recorded
Arrow	fundamental, already in QM via measurement	primitive (P11); reversibility is the coarse-grained shadow
Collapse	real, objective, extra to unitary QM; rejects MWI / decoherence-alone / ensemble	commitment is a real primitive, not derivable from unitary evolution; same three rejections
Spacetime	emergent from a discrete quantum-gravity layer whose collapse drives growth	emergent from the discrete committing substrate

Ellis’s italic hypothesis — collapse as the transition of an indefinite future into a definite past, occurring everywhere continually — is, term for term, the coarse-grained description of P11. What Ellis calls the *crystallizing present*, ED calls the *committing edge*. The tier of this identification is **interpretive and structural**: it is a mapping between two conditional readings of the same quantum and relativistic physics, not a shared theorem. But its significance is not diminished by that. Two derivations that begin as far apart as general relativity and a discrete participation graph, and that arrive at the same non-standard ontology — real flow, open future, fixed past, ontologically special present, objective collapse as the mechanism of becoming — constitute a **consilience** across independent starting assumptions.

4. What ED supplies that Ellis posits: the substrate engine

Ellis’s EBU requires a Scale-0 layer whose collapse actualizes the future at the present-boundary, and asserts that the correct quantum-gravity theory must yield exactly that. Ellis describes the *shape* of this engine — a discrete floor that commits, growing spacetime at the present — and states plainly that he does not have it.

Event Density is a candidate for that engine, and it is *constructed* rather than posited. The committing substrate is the Scale-0 layer; the emergent metric $g \sim 1/b$ and the weak-field Einstein equation are derived from it (Paper_MetricFromTheGraph, Paper_GR-I..IV); the growth of the block is the accumulation of commitments (P11); the “actualization of the future at the present-boundary” is measurement-as-commitment (Report §4). ED even meets Ellis’s stated compatibility class — a discrete, causal, becoming-based substrate in the lineage he names (causal sets, spin foams).

The honest tier: ED is a **conditional structural derivation from thirteen posited primitives**, not a completed quantum gravity, and the full nonlinear Einstein equation is not derived from the primitives (Paper_GR-I delivers the weak field and the khronometric class). (*A separate thermodynamic route, Paper_KhronometricEquationOfState_Jacobson, recovers the Einstein equation in its equation-of-state form with the aether kept among the sources; that is a literature-grounded result, not a from-primitives derivation of the full nonlinear field equations, which remains open.*) So ED does not *close* Ellis’s Scale-0 problem. It offers a specific, constructed candidate for the layer his program requires — which is more than the selection rule (“QG must yield an EBU”) currently provides.

5. The collapse scale Ellis lacks: M_{crit}

Ellis names a viable theory of *when and where* collapse occurs as an outstanding issue — and, importantly, declines a universal collapse scale, treating the trigger as set contextually by the local environment. Event Density goes further, and commits to a scale. If commitment carries a critical multiplicity, there is a quantum-to-classical crossover M_{crit} (Paper_060): a mass above which multi-channel participation can no longer be maintained and the substrate must commit. The associated matter-wave band, $\sim 140\text{--}250$ kDa, is **inherited from empirical extrapolation** (Paper_056), not ED-derived; what ED contributes is the *mechanism* — that a sharp, substrate-forced individuation boundary exists in that regime — and interferometry is currently approaching it.

The tier is explicit: M_{crit} is an **unconfirmed prediction**, not a result. It is a candidate answer to the “when does collapse happen” question Ellis leaves open — and a *bolder* one than he endorsed: where Ellis treats the trigger as contextual with no universal number, ED commits to a universal scale, and is falsifiable for exactly that reason. Where Ellis’s CBU is *designed to accommodate* the existing collapse and delayed-choice experiments (Guerlin et al. 2007; the Kim et al. quantum eraser; Wheeler delayed-choice), ED adds a falsifiable number: a boundary that a static-block or purely-unitary account says should not exist, and that objective collapse says should. This is the one place the two programs, taken together, become sharper than either alone — a shared ontology plus a testable edge that goes beyond Ellis’s own stance.

A note on the CBU’s crinkly present. Ellis’s delayed-choice lag — micro “past” events crystallizing later — has a natural ED reading: commitment occurs at the actual interaction that

fixes a degree of freedom, which can be later than the naïve past-surface. ED reproduces the delayed-choice results (it reproduces the Born rule and no-signaling, A1 capacity-zero), and the interpretive picture — fixing happens at the commit, not before — is the same structure as Ellis’s lagging crystallization surface.

6. Preferred frames: a narrow convergence and a real divergence

The hardest problem for any *becoming* ontology is relativity’s lack of a preferred simultaneity. Both frameworks confront it, and both end up carrying preferred worldlines while treating Lorentz invariance as effective — but they do so in *different theories*, and, as it turns out, they disagree on the direction of the dependence.

Ellis rejects physical realism about Minkowski spacetime (no real vacuum exists; the cosmic background radiation permeates everywhere), so the *real* universe carries geometrically preferred worldlines everywhere: the timelike **Ricci eigenlines**, the local average rest frame of matter. Crucially, these are a foliation *chosen within* the standard Einstein field equations — an ADM gauge / broken solution-symmetry (like the CMB rest frame), **not** a modification of gravity. Ellis posits no dynamical aether field and no Lorentz-violating couplings. The growth is *pointwise along worldlines* by proper time, needing no global simultaneity.

Event Density also carries a preferred structure and also needs no global “now” (commitment is local and pointwise, exactly Ellis’s pointwise-along-worldlines growth). But ED locates it in the **commitment order** — the khronon $u_\mu = \partial_\mu T / |\partial T|$, the arrow made geometric (**Paper_GR-II**) — and this is a *genuine* khronometric (Einstein-aether-class) **modification** of gravity, with computed preferred-frame parameters ($\alpha_2 = 0$ exactly, $\alpha_1 = -4\lambda_{local}$, ~ 70 orders suppressed; **Paper_GR-IV**). ED, not Ellis, lives in the khronometric family.

So the comparison must be stated precisely, because it cuts both ways:

- **The convergence is real but narrow.** Both reject Minkowski realism, both carry geometrically preferred worldlines in the real matter-filled universe, and both keep Lorentz invariance as an *effective* symmetry over a deeper order. That is a genuine structural agreement about what taking becoming seriously requires of the relativistic structure.
- **The theories are not the same.** Ellis stays inside unmodified GR (a preferred foliation); ED is a Lorentz-violating modified gravity (a dynamical khronon). Only ED is in the khronometric / Einstein-aether class (**Paper_KhronometricEquationOfState_Jacobson**, **Paper_GR-IV**).
- **They disagree on the direction of the frame.** ED derives the preferred frame *from* the commitment order (arrow $\rightarrow$ frame). Ellis explicitly argues the reverse — that nothing inherent in collapse supplies a frame, and that the local matter context defines the preferred frame, which then sets the context for collapse (matter $\rightarrow$ frame $\rightarrow$ collapse; *Meshing* §3.2). This is a **substantive disagreement**, not a shared mechanism.

The honest reading, then: a shared *instinct* — that becoming forces preferred worldlines and an effective, emergent Lorentz symmetry — not a shared class and not a shared mechanism. ED goes further than Ellis, into a testable modified-gravity khronon, and in a direction Ellis argued against.

7. Where ED and Ellis diverge — honestly

The convergence is not total, and the differences are real:

- **The Past Hypothesis.** To align the *thermodynamic* arrow with the fundamental direction of time, Ellis retains a fine-tuned low-entropy initial condition (a “past condition”). ED does **not** require it: with irreversibility primitive, the reversible laws are the lossy average and the second law is bookkeeping, so no special initial condition is imported to explain the arrow (Paper_HowTheoryCoarseGrainReality §13). Tier: this ED move is a **reframe** — it identifies low initial entropy with the fewest early commitments (itself a substantive assumption), not a computed derivation of the initial-entropy value — but it is a genuine economy relative to Ellis’s account.
- **The preferred-frame locus** (§6): matter (Ellis) versus the commitment order (ED). Empirically both live in the khronometric class; the difference is which physical structure supplies the foliation.
- **The integrability of the meshing.** Ellis asserts, on observational rather than derivational grounds, that the local proper times mesh into a single growing block (and flags the timelike-crystallization-surface case as unfinished). ED grounds the corresponding fact differently — the emergent metric and its locality come from the graph — but neither framework has a closed proof that the meshing is forced; it is a shared open seam.

8. Honest tiers

- **Collapse-as-present $\approx$ commitment** (§3): *interpretive / structural* — a convergence of two independent readings, not a shared theorem. Load-bearing as consilience, not as proof.
- **ED as the substrate engine** (§4): *conditional structural derivation* from the thirteen primitives; the weak-field metric and khronometric class are delivered, the full nonlinear Einstein equation is not. A constructed candidate for Ellis’s posited Scale-0, not a completed quantum gravity.
- M_{crit} (§5): *unconfirmed prediction*; the ED answer to the “when does collapse happen” question Ellis names as open.
- **Past-Hypothesis dissolution** (§7): *reframe*; an honest economy, not a computed initial-entropy value.
- **Preferred-frame comparison** (§6): *limited structural convergence* — both reject Minkowski realism and carry preferred worldlines, and both treat Lorentz invariance as effective. But only ED is a khronometric modified-gravity theory, and Ellis explicitly locates the frame in matter and argues *against* ED’s arrow-as-frame direction. A shared instinct, not a shared class or a shared mechanism.
- **Convergence, not endorsement:** throughout. Ellis has no knowledge of ED; the agreement is independent arrival.

9. Position statement

Event Density is not a rival to the Evolving Block Universe; it is a candidate **engine** for it. Ellis, from general relativity and the measurement problem, arrives at a growing block whose present is objective wave-function collapse, whose future does not exist, and whose deep basis must be a discrete becoming-substrate he describes but does not build. Among the open problems he

names, two are ones Event Density speaks to directly: that substrate, and a theory of *when* collapse occurs. (Others he lists — the timelike-surface initial-value problem, and the meshing of local proper times into one block — ED does not resolve; see §7.) Event Density, from a discrete committing substrate, arrives at the same ontology — present as the committing edge, open future, fixed past, objective collapse as the arrow — and offers, at their stated tiers, a *constructed* substrate and a *predicted* collapse scale.

The honest claim is therefore narrow and, precisely because it is narrow, strong: two frameworks built from opposite ends of physics converge on a non-standard account of time that most of the field rejects, and the discrete framework supplies two of the pieces the relativistic one names as missing. That is convergence plus candidate completion, not proof — and it is the appropriate relationship between an outsider’s substrate program and one of the field’s most serious defenses of the reality of time.

Cross-references

- **Ellis’s program:** *On the Flow of Time* (Ellis, FQXi 2008); *Time and Spacetime: The Crystallizing Block Universe* (Ellis & Rothman, arXiv:0912.0808); *The Evolving Block Universe and the Meshing Together of Times* (Ellis).
- **Substrate & commitment:** `physics-papers/foundations/Paper_087_13Primitives.md` (P11); Report §4 (measurement as commitment).
- **Emergent metric & gravitational class:** `Paper_MetricFromTheGraph_ForcedTo3D.md`; `Paper_GR-I..IV`; the khronon `Paper_GR-II`; the khronometric couplings `Paper_GR-IV`; the companion `Paper_KhronometricEquationOfState_Jacobson.md`.
- **The arrow as coarse-graining, and the Past-Hypothesis dissolution:** `physics-papers/substrate-evaluation/Paper_HowTheoryCoarseGrainReality.md` (§13); companion `Paper_ConservationColumn.md`.
- **The collapse scale:** `physics-papers/q-compute/Paper_060_Mcrit_Unification.md`; the matter-wave prediction, `physics-papers/predictions/`.
- **No-signalling / commitment-locality:** the A1 capacity-zero result.